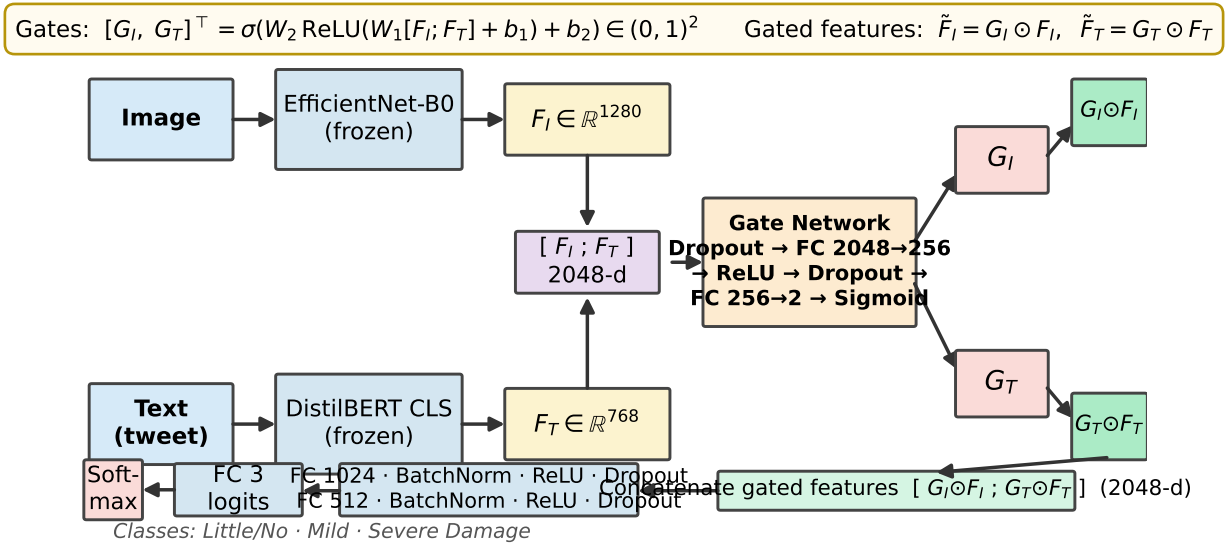
Gates: $[G_I,\ G_T]^\top = \sigma(W_2\,\mathrm{ReLU}(W_1[F_I; F_T] + b_1) + b_2) \in (0,1)^2$
Gated features: $\tilde{F}_I = G_I \odot F_I,\ \ \tilde{F}_T = G_T \odot F_T$
Image
EfficientNet-B0 (frozen)
$F_I \in \mathbb{R}^{1280}$
$[\,F_I\,;\,F_T\,]$ 2048-d
Gate Network
Dropout → FC 2048→256 → ReLU → Dropout → FC 256→2 → Sigmoid
G_I
G_T
$G_I \odot F_I$
$G_T \odot F_T$
Text (tweet)
DistilBERT CLS (frozen)
$F_T \in \mathbb{R}^{768}$
Concatenate gated features $[\,G_I \odot F_I\,;\,G_T \odot F_T\,]$ (2048-d)
FC 1024 · BatchNorm · ReLU · Dropout
FC 512 · BatchNorm · ReLU · Dropout
FC 3 logits
Soft-max
Classes: Little/No · Mild · Severe Damage